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Summary of the book

What Is Consciousness?

4.1 Consciousness Is Limited

Consciousness is our moment-by-moment subjective experiences. At any one time, a person can be conscious of a limited number of things. Change blindness illustrates how selective an individual's attention can be: We often do not notice large changes in an environment because we fail to pay attention.

4.2 Attention Is the Gateway to Conscious Awareness

Attention involves selecting some stimuli for conscious processing while ignoring other stimuli. Attention and consciousness go hand in hand. By filtering out some information, attention helps determine what enters conscious awareness.

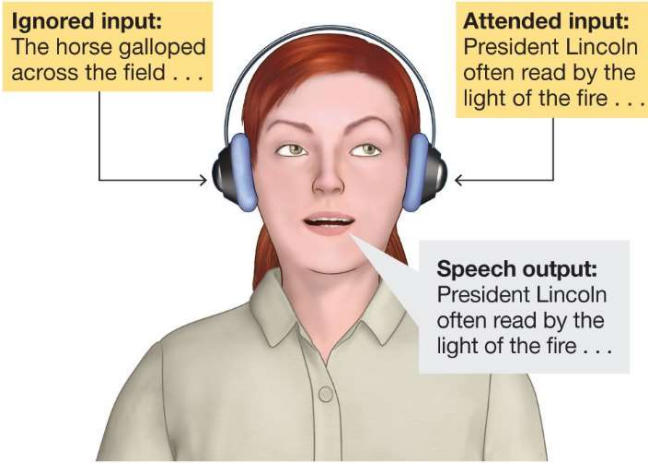


FIGURE 4.4
Shadowing
In this procedure, the participant receives a different auditory message in each ear. The participant is required to repeat, or "shadow," only one of the messages.

4.3 Laptops in the Classroom

Because consciousness is limited, attending to a laptop during class may make a student more likely to miss the content of a lecture.

4.4 Unconscious Processing Can Influence Behavior

Thought and behavior can be influenced by stimuli that are not experienced at a conscious level, as demonstrated by implicit biases and priming. Subliminal perception occurs when people are influenced by hidden messages. Although stimuli can have some effects on people without their awareness, there is currently no evidence that subliminal messages can compel people to perform complex actions against their will.

4.5 Automatic Processing Reduces Demands on Consciousness

Although learning a task may require a great deal of concentration and controlled processing, highly practiced tasks can become automatic and require less conscious effort. Reducing the demand for conscious awareness can be a benefit in that it frees up consciousness for other tasks, but automaticity can also have costs depending on the situation.

What Is Altered Consciousness?

4.6 Meditation Produces Relaxation by Guiding Attention

By guiding attention, meditation can produce an altered state of consciousness. The goal of meditation, particularly as it is practiced in the West, is to bring about a state of deep relaxation. Studies suggest that meditation can have multiple benefits for people's physical and mental health.

4.7 You Be the Psychologist: Does Meditation Change the Structure of the Brain?

There are many challenges in designing studies that can conclusively demonstrate that meditation reduces brain changes associated with aging.

4.8 People Can Lose Themselves in Activities

Exercise, religious practices, and other engaging activities can produce a state of altered consciousness called flow. In this state, people become completely absorbed in what they are doing. Flow is experienced as a positive state. In contrast to activities that generate flow, activities used to escape the self or reduce self-awareness can have harmful consequences.

4.9 Hypnosis Is Induced Through Suggestion

Scientists debate whether hypnotized people merely play the role they are expected to play or truly experience an altered state of consciousness. Consistent with the latter view, brain imaging research has shown changes in brain activity among hypnotized participants.

What Is Sleep?

4.10 Sleep Is an Altered State of Consciousness

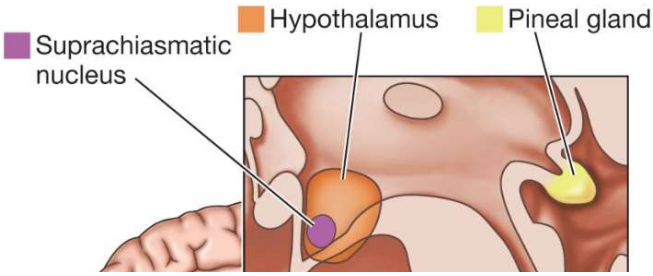
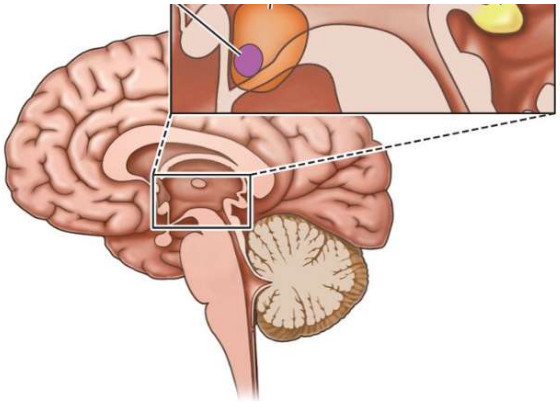


FIGURE 4.15
The Pineal Gland and the Sleep/Wake Cycle
Changes in light register in the suprachiasmatic nucleus of the hypothalamus. In response, this region signals the pineal gland when the time for sleep or the



suprachiasmatic nucleus of the hypothalamus. In response, this region signals the pineal gland when the time for sleep or the time for wakefulness has come.

Sleep is characterized by four stages in which brain activity varies. These stages range from short bursts of irregular waves (stages 1–2) to large, slow brain waves during deep, restful sleep (stages 3–4). REM sleep is marked by a return to short, fast brain waves and is accompanied by rapid eye movements, body paralysis, and dreaming.

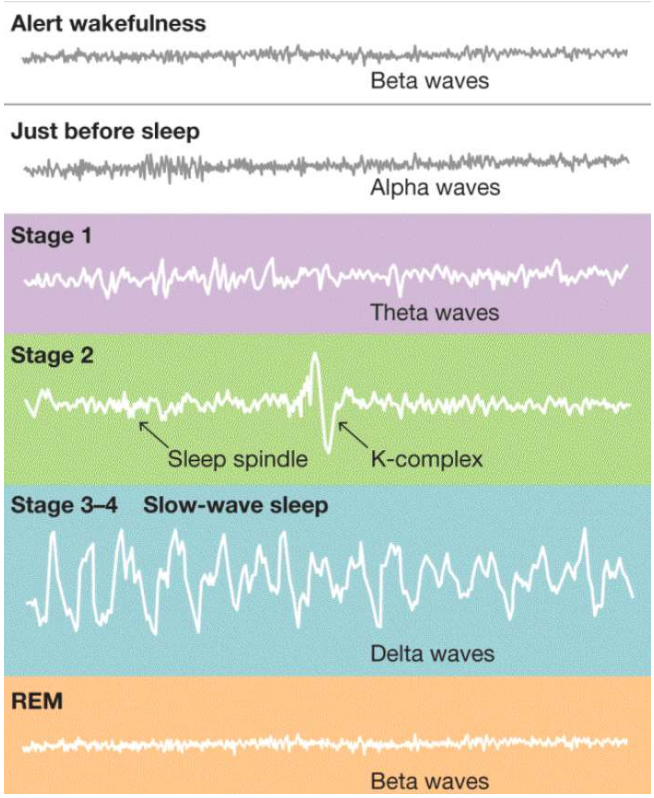


FIGURE 4.16
Brain Activity During Sleep
Using an EEG, researchers can measure the patterns of electrical brain activity during different stages of normal sleep.

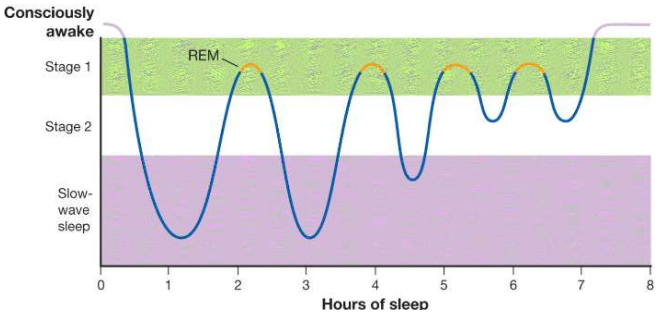


FIGURE 4.17
Stages of Sleep
This chart illustrates the normal stages of sleep over the course of the night.

4.11 People Dream While Sleeping

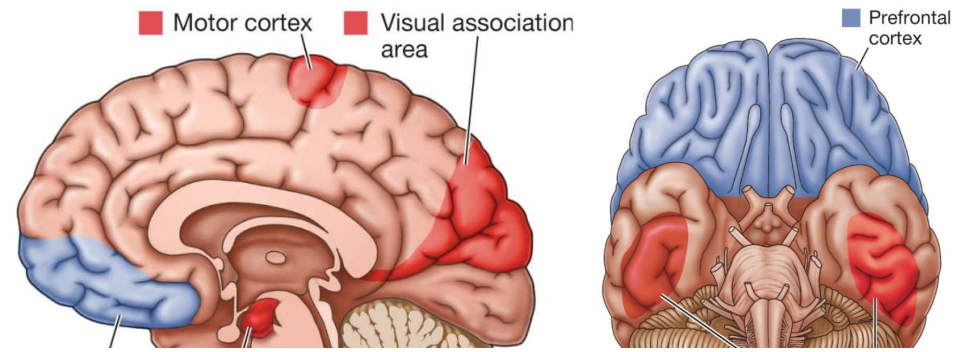
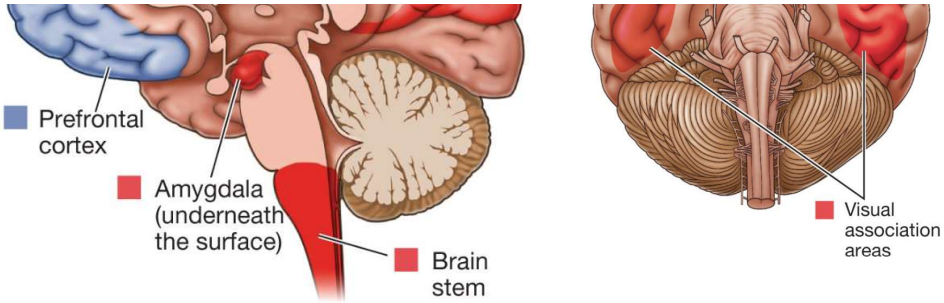


FIGURE 4.18
Brain Regions and REM Dreams
These two views of the brain show the regions that are activated (shown in red) and deactivated (shown in blue) during REM sleep. (a) As seen here from the side, the motor cortex, brain stem, and visual association areas are activated, as are brain regions involved in motivation, emotion, and reward (e.g., the amygdala). A region of the prefrontal cortex is deactivated. (b) As shown here from



areas are activated, as are brain regions involved in motivation, emotion, and reward (e.g., the amygdala). A region of the prefrontal cortex is deactivated. (b) As shown here from below, other visual association areas are activated as well. (This view also reveals the full size of the prefrontal cortex.)

REM dreams and non-REM dreams activate and deactivate distinct brain regions. Sigmund Freud believed that dreams reveal unconscious conflicts, but evidence does not support this view. Activation-synthesis hypothesis posits that dreams are the product of the mind's efforts to make sense of random brain activity during sleep.

4.12 Sleep Is an Adaptive Behavior

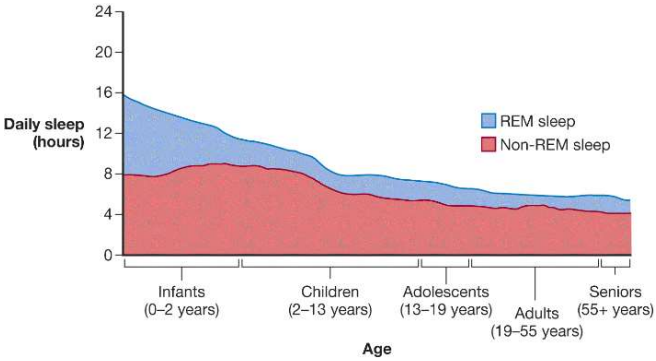


FIGURE 4.19
Sleep Patterns Across the Life Span
As we age, the amount of sleep required decreases and less time is spent in REM sleep.

Three theories account for sleep: Sleep allows the body, including the brain, to rest and restore itself. Sleep protects animals from harm at times of the day when they are most susceptible to danger. And sleep facilitates learning through the strengthening of neural connections.

4.13 Sleep Disorders Interfere with Daily Life

Insomnia is an inability to sleep that causes significant problems in daily living. Other sleep disorders include obstructive sleep apnea, narcolepsy, REM behavior disorder, and somnambulism (sleepwalking).

4.14 Psychology Outside the Lab: How Can You Get a Good Night's Sleep?

Good sleep hygiene involves planning, setting priorities, sticking to the plan, and saying no to behaviors that might interfere with sleep. To sleep well, people should have a routine, avoid stimulants or alcohol before trying to sleep, exercise during the day, reserve the bed for sleep, relax at bedtime, get up for a while if they have trouble falling asleep, avoid electronic devices late at night, and not try to catch up on lost sleep.

How Do Brain Injury and Drugs Affect Consciousness?

4.15 Brain Injury Can Diminish Consciousness

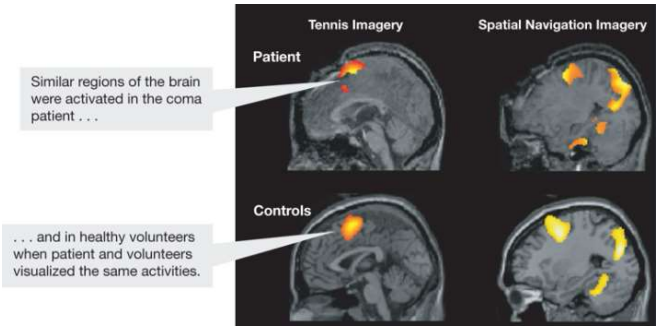


FIGURE 4.24
In a Coma but Aware
The brain images on the top are from the patient, a young woman in a coma who showed no outward signs of awareness. The images on the bottom are a composite from the control group, which consisted of healthy volunteers. Both the patient and the control group were told to visualize, first, playing tennis, and second, walking around a house. Right after the directions were given, the neural activity in the patient's brain appeared similar to the neural activity in the control group's brains.

Concussions can temporarily alter consciousness. A person in a minimally conscious state shows some awareness of external stimuli. A person with unresponsive wakefulness syndrome does not show consciousness. A person who is brain dead has no brain activity and is not alive; the person's body is being kept alive artificially.

4.16 Drugs Alter Consciousness by Changing Brain Neurochemistry

TYPE	PSYCHOLOGICAL EFFECT(S)	EXAMPLES	NEUROTRANSMITTER SYSTEM(S)
Stimulants	Increase behavioral and mental activity	Amphetamines, methamphetamine, cocaine, nicotine, caffeine	Dopamine, norepinephrine, acetylcholine (nicotine)
Depressants	Decrease behavioral and mental activity	Antianxiety drugs (barbiturates, benzodiazepines), alcohol	GABA
Opioids	Reduce the experience of pain, bring pleasure	Heroin, morphine, codeine	Endorphins
Hallucinogens/psychedelics	Alter thoughts or perceptions	LSD, PCP, peyote, psilocybin mushrooms	Serotonin (LSD, peyote, psilocybin), glutamate (PCP)
Combination	Mixed effects	Marijuana, MDMA	Cannabinoid (marijuana), serotonin, dopamine,

			norepinephrine (MDMA)

Psychoactive drugs are mind-altering substances. Each drug affects one or more specific neurotransmitters, leading to psychological effects. They can be divided into categories
(stimulants, depressants, opioids, hallucinogens) based on these effects, but some psychoactive drugs do not fit neatly into categories because they have various effects.

4.17 People Use—and Abuse—Many Psychoactive Drugs

Stimulants, including cocaine and amphetamines, increase behavioral and mental activity. Opioids, including morphine and heroin, produce a relaxed state, insensitivity to pain, and euphoria. During the past decade, there has been an epidemic of opioid abuse, especially of heroin addiction. THC (the active ingredient in marijuana) produces a relaxed state, an uplifted mood, and perceptual and cognitive distortions. MDMA, or ecstasy, produces energizing and hallucinogenic effects.

4.18 Alcohol Abuse Is Responsible for Many Societal Problems

Alcohol has its effects by activating GABA receptors. As a depressant, it decreases behavioral and mental activity. Throughout the world, men drink more than women. Alcohol's effects are influenced by the drinker's expectations.